TiHUB-01

4-Port Serial-to-Ethernet Modbus Gateway

User Manual

v0.1.0

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1 General Introduction

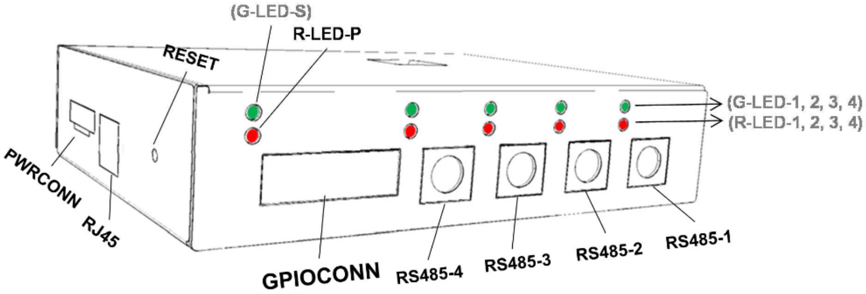
TiHUB-01 is a 4-Port Modbus gateway that can bridge up to 4 TiACC sensor nodes to Ethernet. The bridging is accomplished by conversion between Modbus-RTU and Modbus-TCP communication protocols. In such task TiHUB-01 serves as a Modbus-TCP Server (Slave) that receives request from user Modbus-TCP Client (Master) and convert it to Modbus-RTU request to send to connected TiACC sensor node. In similar way TiHUB-01 get the Modbus-RTU response from TiACC sensor node and convert it to Modbus-TCP response to user Master. TiHUB-01 has 4 serial ports that each dedicate a high speed Modbus-RTU connection to a TiACC sensor node. It has one GPIO connection. The TiHUB-01 itself is also a Modbus-TCP Server (Slave) that is configurable by user Modbus-TCP Client (Master) through registers. It is solidly built to work in a harsh environment with IP67 rating.

1.1 Features

- Bridge between Modbus-RTU and -TCP protocols
- 4 Serial ports:
  - RS-485 interface
  - Baud rate up to 4 MHz
  - Modbus-RTU Master
- 1 Ethernet ports:
  - 10/100 BaseTX Ethernet
  - 1 Modbus-TCP Server (Slave) for gateway configuration
  - 4 Modbus-TCP Server (Slave) for sensor node bridge
  - 4 TCP Server for sensor data stream
- Single +24V operation voltage
- 4x +5V power out to TiACC nodes
- -40°C to +85°C operation temperature
- Configuration options: web console and Modbus-TCP Server (Slave)
- Embedded NVM for data logging
- RoHS-compliance aluminum case:
  - 4x female Min-Din-6 for serial port and +5V power out to TiACC nodes
  - 1x GPIOCONN with 8-Input / 8-Output
  - 1x RJ45 for Ethernet connection
  - 1x PWRCONN for +24V power in
  - 1x RESET push button
  - 1x Power LED
  - 150x110x35 mm^3

1.2 Applications

Serial-to-Ethernet Modbus Gateway for TiACC sensor node



1.3 Specifications

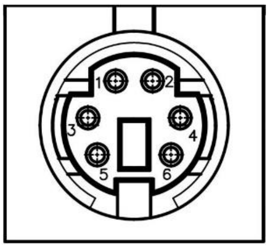


Figure 1 RS485 Connector

Table 1 RS485-x, x=1~4, Pin Descriptions

Pin#	Name	Type	Description
1	B	Bus I/O	RS-485 I/O port B (complementary to A)
2	A	Bus I/O	RS-485 I/O port A (complementary to B)
3	GND	Power	Ground pin
4	VDD5V	Power	+5V operation voltage power out
5	RST	Digital I/O	Configurable as reset pin (digital output)or sensor detect pin (digital input)
6	INT/TRIG	Digital I/O	Configurable as interrupt pin (digitalinput) or trigger pin (digital output)

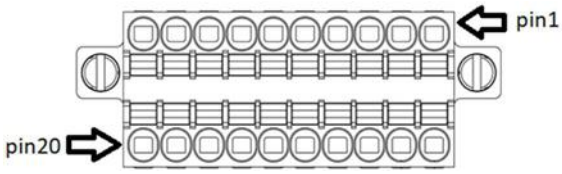


Figure 2 GPIOCONN Connector

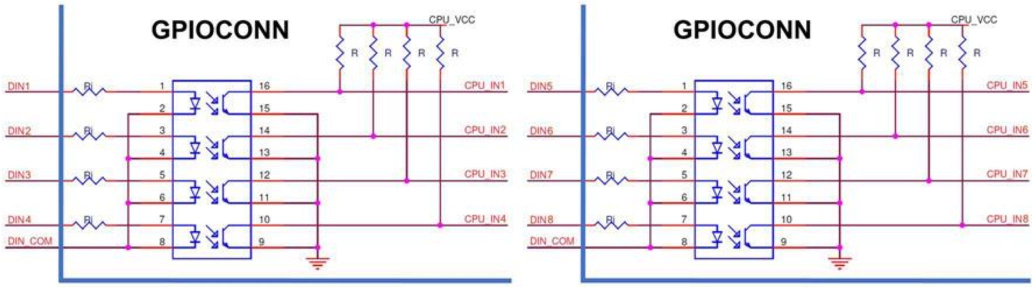


Figure 3 GPIOCONN Input Pin Schematic

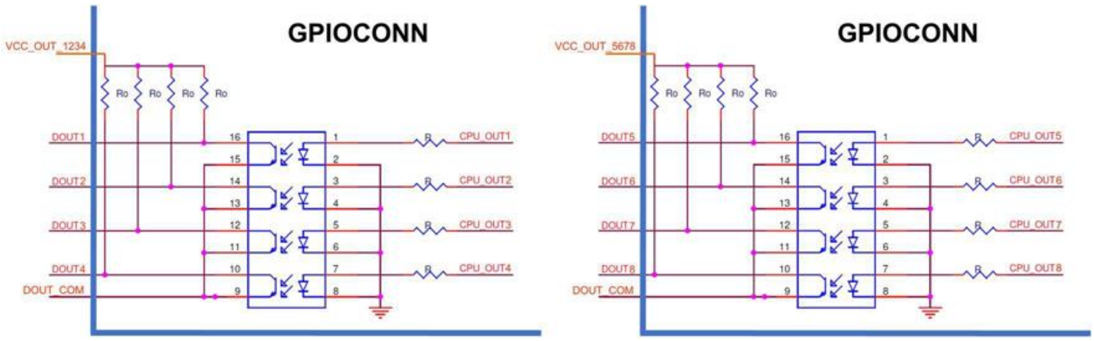


Figure 4 GPIOCONN Output Pin Schematic

Table 2 GPIOCONN Pin Descriptions

Pin#	Name	Type	Description
1	VCC_OUT_1234	Power	VCC for DOUT1~4
2	VCC_OUT_5678	Power	VCC for DOUT5~8
3	DOUT1	Digital Output	Digital output 1
4	DOUT2	Digital Output	Digital output 2
5	DOUT3	Digital Output	Digital output 3
6	DOUT4	Digital Output	Digital output 4
7	DOUT5	Digital Output	Digital output 5
8	DOU6	Digital Output	Digital output 6
9	DOUT7	Digital Output	Digital output 7
10	DOUT8	Digital Output	Digital output 8
11	DIN1	Digital Input	Digital input 1
12	DIN2	Digital Input	Digital input 2

13	DIN3	Digital Input	Digital input 3
14	DIN4	Digital Input	Digital input 4
15	DIN5	Digital Input	Digital input 5
16	DIN6	Digital Input	Digital input 6
17	DIN7	Digital Input	Digital input 7
18	DIN8	Digital Input	Digital input 8
19	DOUT_COM	Power	Common ground for DOUTx pins
20	DIN_COM	Power	Common ground for DINx pins

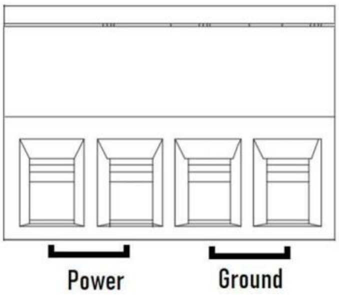


Figure 5 PWRCONN Connector

Table 3 PWRCONN Pin Descriptions

Pin Name	Type	Description
Power	Power	TiHUB-01 operation voltage VDD24V in
Ground	Ground	TiHUB-01 ground

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Table 4 Generation Specification

Environment temperature Ta = 25°C if not specified otherwise

Parameter	Conditions	Min.	Typ.	Max.	Unit
Operation voltage (VDD24V)	T_a = -40°C ~ +85°C	—	24.0	—	V
Operating current	Gateway itself only, excluding sensor power	—	0.5	—	A
NVM for data logging		—	27	—	GB
Operation temperature		-40	—	+85	°C
Storage temperature		-40	—	+125	°C
Ethernet Interface					
10/100Base TX Port	×1				
Magnetic isolation protection	1.5kV				
Connector	RJ45				

Ethernet Services

Service Name	Port Number	Description
MBAP, Gateway	502	Gateway Modbus-TCP Server (Slave)
MBAP, SN1	505	Sensor node SN1 Modbus-TCP Server (Slave)
MBAP, SN2	506	Sensor node SN2 Modbus-TCP Server (Slave)
MBAP, SN3	507	Sensor node SN3 Modbus-TCP Server (Slave)
MBAP, SN4	508	Sensor node SN4 Modbus-TCP Server (Slave)
SN1 Data Stream	1505	Sensor node SN1 data stream TCP Server
SN2 Data Stream	1506	Sensor node SN2 data stream TCP Server
SN3 Data Stream	1507	Sensor node SN3 data stream TCP Server
SN4 Data Stream	1508	Sensor node SN4 data stream TCP Server

Ethernet Software Features	
Configuration Options	Web console, Modbus-TCP Server (Slave)
Management	ARP, DHCP client, DNS, HTTP, TCP/IP, NTP client, mDNS, FTP
Time Management	NTP Client
Max. Client Connections	Port 502, 505, 506, 507, 508: No limit Port 1505, 1506, 1507, 1508: 5
Serial Interface	
No. of Ports	4
Connector	Female Min-Din-6, with +5V power out

Baud rate	Up to 4MHz
Data bits	8
Parity	None
Stop bits	1
Flow control	None
RS-485-2w	Data+, Data-, GND
Terminator for RS-485	120 ohms
Industry protocol	Modbus-RTU Master
LED	
R-LED-P	Red power LED. Light-up when VDD24V connect to +24V
G-LED-S	Green status LED. Not used.
G-LED-1~4	Green sensor node status LEDs. Not used.
R-LED-1~4	Red sensor node status LEDs. Not used
Button	
RESET	Reset push button. Push RESET to reset TiHUB-01.

Table 5 RS-485 Bus I/O Characteristics

Parameter	Symbol	Pin	Note	Min.	Typ.	Max.	Unit
Differential input voltage	VID	A/B		-15	0	15	V
Positive-going inputthreshold voltage	VTH+	A/B		Note	-100	-20	mV
Negative-going inputthreshold voltage	VTH-	A/B		-200	-130	-1	mV
Input hysteresis	VHYS	A/B		—	30	—	mV
Output high voltage	VOH	A/B		VCC-0.6	VCC-0.4	—	V
Output low voltage	VOL	A/B		—	0.4	0.6	V

Note: Under any circumstances, VTH+ is guaranteed to be at least VHYS higher than VTH–.

Table 6 GPIO Characteristics

Parameter	Symbol	Pin	Min	Typ	Max.	Unit
Input Voltage	VIN	DIN×	3.3	5	5.5	V
Input Serial Resistance	Ri	DIN×	—	390	—	Ω
Input Reverse Voltage	VREV	DIN×	—	—	6	V
Output Voltage	VOUT	VCC_OUT_1234 VCC_OUT_5678	3.3	5	5.5	V
Output Current	IOUT	DOUTx	15	20	25	mA
Output Pull-up	Ro	DOUTx	—	470	—	Ω
Collector-Emitter Breakdown Voltage	BVCEO	DOUTx DOUT_COM	80	—	—	V
Emitter-Collector Breakdown Voltage	BVECO	DOUTx DOUT_COM	7	—	—	V

2 Block Diagram and Connection

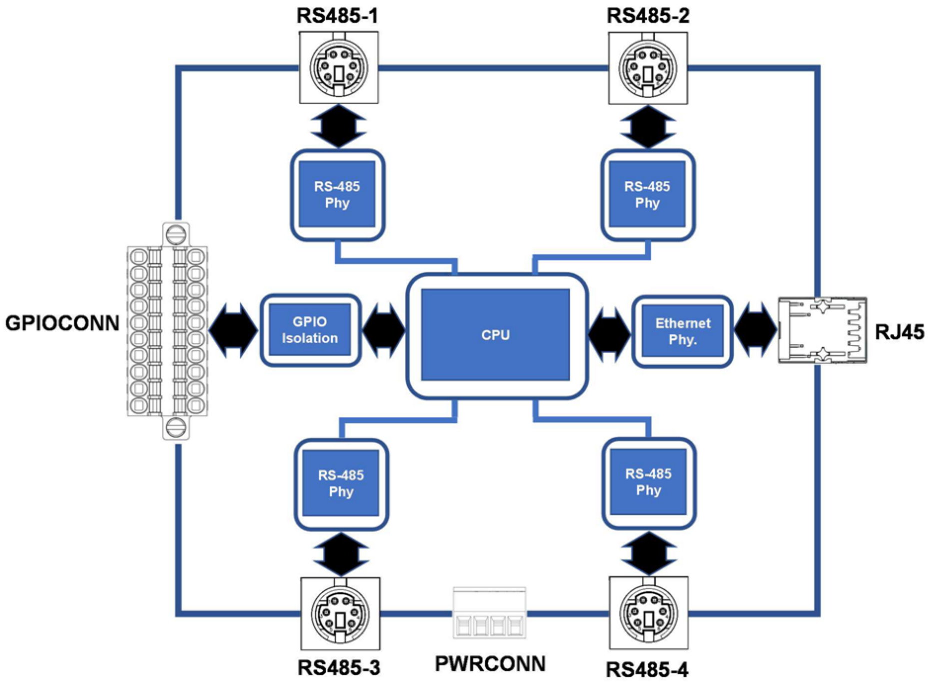

Figure 6 TiHUB-01 Block Diagram

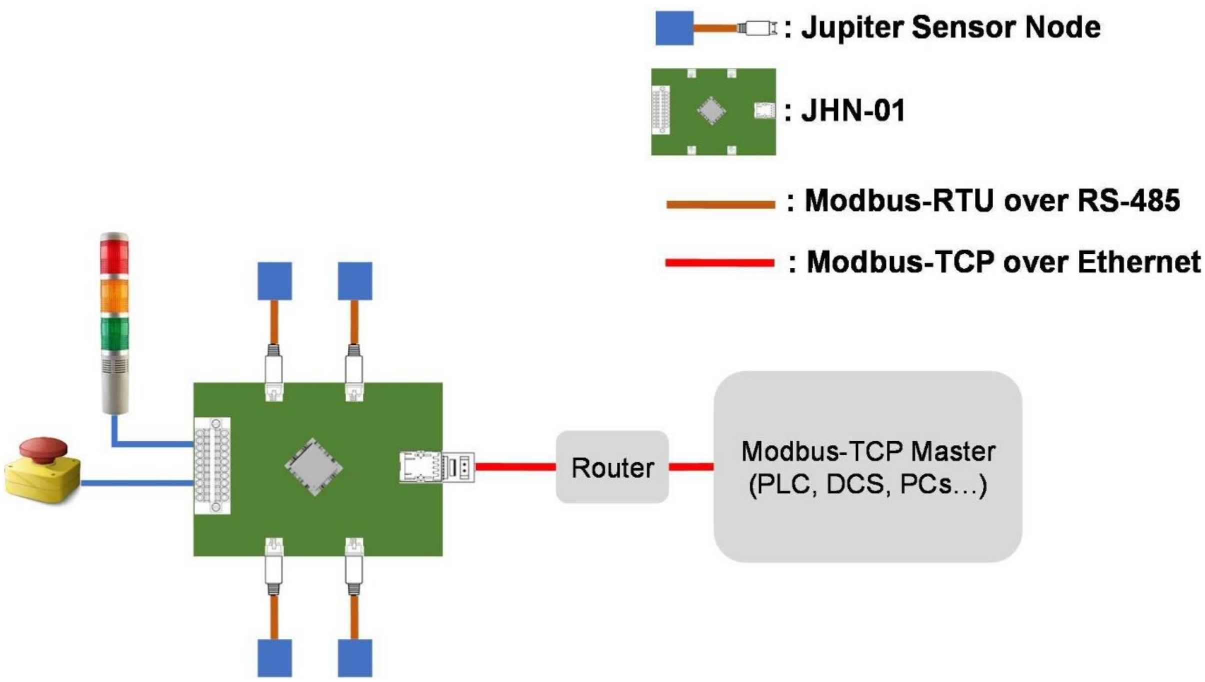

Figure 7 TiHUB-01 Connection Example

3 Function Description

3.1 Start-up Behavior

The red power LED (R-LED-P) will light up and TiHUB-01 will power-up when +24V operation voltageis connected.
Upon power-up, TiHUB-01 will establish network and start the following TCP services for users:

- One gateway configuration service
- Four sensor node bridging services
- Four sensor data stream services

By default TiHUB-01 just starts these services and wait for requests from user Master for further action. User may change from this start-up to idle behavior to automatic data servers or loggers. Please refer to the below section, “Automatic Sensor Node Data Stream Server and Logger”, for details.

3.2 Network

In the start-up, TiHUB-01 will try to establish network by first requesting from the DHCP server an IP address and other parameter values for accessing the network services. If for any reason this attempt fails, TiHUB-01 will try to assign to the default static IP address “169.254.7.10”. If IP addressconflict occurs, TiHUB-01 will try to assign to other address in the 169.254.*.* range.

Access to TiHUB-01 is intuitive with TiHUB-01’s mDNS (Multicast DNS) support. The hostname, “cmc- sensor-hub.local”, will be broadcast on any network when TiHUB-01 attaches the network. User can use the command, “ping cmc-sensor-hub.local”, to elicit response from TiHUB-01

3.3 Automatic Sensor Node Data Stream Server and Logger

TiHUB-01 may be configured to become automatic sensor data stream server and logger after start-up. Below summarize the steps for such configuration. Note these setting will be saved in the NVM in TiHUB-01 and become effective for the next start-up.

The following steps are carried out on the gateway configuration service (port 502).

1. Set STARTODR1~4 (52h~55h) registers to desired automatic start-up sensor ODR values.
2. If data-logging is required, set AUTOLOG (51h) register to 0x01, else set AUTOLOG (51h) register to 0x00. Please refer to STRMLOG (2Ch) register description for the log file format definition.
3. Set STARTMOD (50h) register to 0x01 to enable automatic data stream server/logger.
4. Restart TiHUB-01 by power-down and power-up again. After start-up,TiHUB-01 will automatically stream all connected sensor node data to respective port (port 1505~1508), and log the data if AUTOLOG is set.
5. User may selectively set STEN_Sx, x=1~4 (20h[4:0]) bit to 1’b0 to turn off the sensor data stream any time at will.

3.4 Access to TiACC Sensor Node

TiHUB-01 can bridge up to 4 TiACC sensor nodes to Ethernet.The sensor nodes are exposed to user by the sensor no de bridging services(port 505~508).

The bridging is accomplished by conversion between Modbus-RTU and Modbus-TCP communication protocols. In such task TiHUB-01 serves as a Modbus-TCP Server (Slave) that receives request from user Modbus-TCP Client (Master) and convert it to Modbus-RTU request to send to connected TiACC sensor node. In similar way TiHUB-01 get the Modbus-RTU response from TiACC sensor node and convert it to Modbus-TCP response to user Master. Please refer to “Modbus Interface” section for more detail information.

3.5 Sensor Node Data Stream Server and Logger

In addition to use Modbus-TCP to request sensor node data, TiHUB-01 can be configured to sensor node data stream servers through port 1505~1508. <u>If such stream mode is enabled, the device will continuously stream 3-axis acceleration data to the stream port. This stream behavior violates the pure master/slave scheme and thus is not compatible to Modbus standard.</u>

Below summarized the recommended steps to setup the data stream. The following steps are carried out on the gateway configuration service (port 502).

1. Set the sensor ODRs by setting 21h~24h registers. For example if data rate 6400Hz is desired for SN1, set 21h = 0x1900.

2. Set the Sensor Stream FIFO Buffer Length registers (26h~2Ah) to 0x40.
3. If data logging is required, set 2Ch = 0x01 to enable sensor data stream log function.
4. Turn on sensor node power by set PW_Sx (2Dh[4:0]) bits to 1'b1 for SNx, x=1~4 respectively.
5. Set STEN_Sx, x=1~4 (20h[4:0]) to 1'b1 to enter the data stream mode. User may selectively set STEN_Sx to 1'b0 to turn off the sensor data stream any time at will.
6. User now can continuously get the data stream through port 1505~1508. Please refer to 20h register description for the data stream format definition.
7. If data logging is enabled, user can use FTP, "ftp cmc-sensor-hub.local", to access the logging file. Please refer to 2Ch register description for log file format definition.

3.6 Access GPIO

TiHUB-01 provide a set of 8+8 general purpose GPIO for user-defined function. Access to the GPIO is accomplished through the gateway configuration service (Gateway MBAP on port 502).

The GPIOCONN's DOUT1~8 High/Low status is mapped to the ON/OFF status of discrete coils of #0 to #7 respectively. User can use Modbus function 01/05/15 to read/write the status of the output pin.

The GPIOCONN's DIN1~8 High/Low status is mapped to ON/OFF status of discrete input of #0 to #7 respectively. User can use Modbus function 02 to read the status of the input pin.

For details please refer to the Modbus Interface section.

4 User Registers

4.1 User Register Map

Modbus Addr	Register Addr.	Name	Access	Default
40001	00h	Product Identification	R	0x0C01
40002	01h	Device Identification	R	0x00C9
40007	06h	Sensor Connection Status	R	0x0000
40033	20h	Sensor Stream Mode Enable	R/W	0x0000
40034~40037	21h~24h	Sensor Stream Output Data Rate (ODR) Configuration	R/W	0x0032
40039~40042	26h~2Ah	Sensor Stream FIFO Buffer Length	R/W	0x0040
40044	2Bh	Sensor Stream Connection Count	R	0x0000
40045	2Ch	Sensor Stream Auto-Log Configuration	R/W	0x0000
40046	2Dh	Sensor Power Control	RW	0x001F
40049~40053	30h~34h	Sensor Stream CRC Error Count	RW	0x0000
40081	50h	Start-up Mode Configuration	RW	0x0000
40082	51h	STARTUP_AUTO_LOG	RW	0x0000
40083~40086	52h~55h	STARTUP_ODR1~4	RW	0x1900
40097~40100	60h~64h	Sensor Modbus-TCP Port	R	0x01F9 0x01FA 0x01FB 0x01FC 0x01FD
40102~40105	65h~68h	Sensor Stream Port	R	0x05E1 0x05E2 0x05E3 0x05E4
40161~40164	A0h~A3h	Version	R	TBD
40177~40180	B0h~B3h	Sensor SN1 FW Version	R	N/A
40182~40185	B5h~B8h	Sensor SN2 FW version	R	N/A
40187~40190	BFh~BDh	Sensor SN3 FW version	R	N/A
40192~40195	BFh~C2h	Sensor SN4 FW Version	R	N/A

4.2 Description of Registers

ModbusAddr: 40001, RegisterAddr: 00h

Product Identification Registers

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
PID								R	0x0C

PID: part ID, fixed to 0x0C

bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0	Access	Default
INFO								R	0x01

INFO: part info, fixed to 0x01

ModbusAddr: 40002, RegisterAddr: 01h

Device Identification Registers

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
DID								R	0x00C9

DID: device ID, fixed to 0x00C9

ModbusAddr: 40007, RegisterAddr: 06h

Sensor Connection Status Register

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
———								R	0x00

bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0	Access	Default
———				S4_ST	S3_ST	S2_ST	S1_ST	R	0x00

SNx_ST: connection status bit for sensor SNx, x=1~4

- 1'b1: sensor SNx is connected
- 1'b0: sensor SNx is not connected

ModbusAddr: 40033, RegisterAddr: 20h

Sensor Stream Mode Enable Register

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
———								RW	0x00

bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0	Access	Default
———				STEN_S4	STEN_S3	STEN_S2	STEN_S1	RW	0x00

STEN_Sx: Stream mode enable bit for sensor SNx, x=1~4

- 1'b1: enable sensor SNx stream mode
- 1'b0: disenable sensor SNx stream mode

<u>If the stream mode is enabled, the device will continuously stream 3-axis acceleration data to thestream port. This stream behavior violates the pure master/slave scheme and thus is not compatibleto Modbus standard.</u>

<u>Data Stream</u>

The data stream packet format is summarized in the table below.

Byte	1	2	3	…	2xan	1+2xan	…	2xanxn	1+2xanxn
Name	ADDR	AXIS1_1H	AXIS1_1L	…	AXIS{an}_1H	AXIS{an}_1L	…	AXIS{an}_{n}H	AXIS{an}_{n}L

ADDR: sensor SNx slave address, i.e. 0x43 for TiACC-01

an: total number of the enabled axes. Fixed to be 3

n: number of set of buffered data configured by STRMLENx register

AXIS{x}_{y}H: high byte value of the x^{th} selected axis of the y^{th} buffered data set,x = 1~an, y=1~n

AXIS{x}_{y}L: low byte value of the x^{th} selected axis of the y^{th} buffered data set,x = 1~an, y=1~n

<u>ModbusAddr: 40034~37, RegisterAddr: 21h~24h</u>

Sensor Stream Output Data Rate (ODR) Configuration Registers

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_ODR1~4								RW	0x0032

STRMODRx, x=1~4 registers configure the sensor SNx stream mode ODR in Hz. The default valueis 0x0032, i.e. 50Hz. Sensor SNx may have its own limit on the ODR setting value. User should refer to the datasheet of the connected sensor for such information

ModbusAddr: 40039~42, RegisterAddr: 26h~2Ah

Sensor Stream FIFO Buffer Length

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_LEN1~4								RW	0x0040

STRMLENx, x=1~4 registers configure the FIFO buffer length for the sensor SNx stream. Each stream packet of sensor SNx will have STRMLENx sets of data.

ModbusAddr: 40044, RegisterAddr: 2Bh

Sensor Stream Connection Count

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_CONN_COUNT								R	0x0000

STRMCONN register contain the total number of stream port connection. TiHUB-01 has 4 stream port,with each permitting maximum 5 connection. So the maximum register value is 20.

ModbusAddr: 40045, RegisterAddr: 2Ch

Sensor Stream Auto-Log Configuration

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_AUTO_LOG								RW	0x0000

STRMLOG register control the auto data logging function in the stream mode.
STRM_AUTO_LOG[15:0]:

- 0x00: disenable data logging
- 0x01: enable data logging

When the auto data logging is enabled, the data stream will be saved in the non-volatile memory within TiHUB-01. User can use the FTP to fetch the saved data for further analysis. If the memory isfull and the logging is still on, the oldest log file will be automatically overwritten.

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The data log files are placed within respective directory “Sensor_x” for data from sensor node SNx, x=1~4, as summarized in the following table

Sensor Node	Stream Port	Log File Directory
SN1	1505	Sensor_1
SN2	1506	Sensor_2
SN3	1507	Sensor_3
SN4	1508	Sensor_4

The sensor data are stored in *.bin files within respective directory “Sensor_x”. Each *.bin file canstore up to 60 sec of raw data using binary format. If the log duration is longer than 60 sec, the logging will split into multiple *.bin files each having 60-sec data.

The *.bin file starts with a 20-byte header as described below

Field	Magic Number				Data Type				Sampling Rate				Start Date-Time							
Byte	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

- Magic Number: Int32, fixed to be 0x288F75EA
- Data Type: Int32, value can be 1~4. Represent the stored data format.
  - 1: data is in 2-byte Float16 format
  - 2: data is in 4-byte Float32 format
  - 3: data is in 2-byte Int16 format
  - 4: data is in 4-byte Int32 format
- Sampling Rate: Int32, the data sampling rate in Hz.
- Start Date-Time: Int64, the start recording time of this file. The value is the epoch-like time, the number of nanoseconds that have elapsed since January 1, 1970

Following the header are sensor data section. The sensor data section contains 60x “Sampling Rate”sets of data. Each set have 3-axis data in the order of X-, Y- and Z-axis. Totally the sensor data section will have 60x “Sampling Rate” x3 data. The data type is specified in the “Data Type” field of the header. Depending on the data type, the sensor data section will have 3600 or 7200x “Sampling Rate” bytes of data for 2- or 4-byte data length respectively.

For “Data Type” of 1 or 3, individual data length is 2-byte, the sensor data section will look like

Field	X1		Y1		Z1		X2		Y2		Z2		X3		Y3		Z3		X4		Y4		Z4		…
Byte	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	…

For “Data Type” of 2 or 4, individual data length is 4-byte, the sensor data section will look like

Field	X1				Y1				Z1				X2				Y2				Z2				…
Byte	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	…

ModbusAddr: 40046, RegisterAddr: 2Dh

Sensor Power Control

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
———								RW	0x00

bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0	Access	Default
———				PW_S4	PW_S3	PW_S2	PW_S1	RW	0x1F

PW_Sx: power control for sensor node SNx, x=1~4

- 1’b1: +5V power is enabled for sensor node SNx
- 1’b0: +5V power is disenabled for sensor node SNx

+5V power is enabled for all sensor nodes by default. This default power-on state is necessary for the correct functioning of sensor node hot-plugging. Set this bit off when this hot-plug behavior isnot anticipated. NOTE: turn off all the sensor stream before hot-plugging any sensor node.

ModbusAddr: 40049~53, RegisterAddr: 30h~34h

Sensor Stream CRC Error Count

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_CRC1~4								RW	0x0000

STRMCRCx, x=1~4 registers accumulate the CRC error count from sensor node SNx, x=1~4. User should clear these registers before turn on stream mode to correctly monitor the data stream quality of current session. If the CRC error count increases dramatically, user should check thesensor node connection to diminish the CRC error.

In stream mode, each enabled sensor node will continuously stream data to TiHUB-01.The data stream packet of sensor SNx contains STRMLENx (26h~2Ah) sets of data with CRC appended in the end. TiHUB-01 will calculate the CRC of the received data packet. If the calculated and received CRC values are different, the STRMCRCx register value will increase by one.

ModbusAddr: 40081, RegisterAddr: 50h

Start-up Mode Configuration

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STARTUP_MODE								RW	0x0000

STARTMOD, AUTOLOG and STARTODRx, x=1~4 registers configure the start-up (power-up) behavior of the TiHUB-01. These register values are kept in the NVM and become effective for the next power-up.

STARTUP_MODE[15:0]:

- 0x00: default Modbus mode. TiHUB-01 will power-up to wait for command from Modbus master for further action.
- 0x01: data stream mode. TiHUB-01 will power-up to stream data from all connected sensor nodes with ODR set by STARTODRx (52h~55h) registers respectively. User may listen to respective stream port to receive sensor data from the data stream. Please refer to the STRMCFG (20h) register for the stream data format.

ModbusAddr: 40082, RegisterAddr: 51h

Start-up Mode Configuration

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STARTUP_AUTO_LOG								RW	0x0000

STARTODRx, x=1~4 contains the ODR setting for sensor node SNx when start-up to data stream mode is enabled (STARTUP_MODE[15:0] = 0x01).
STARTUP_AUTO_LOG[15:0]:

- 0x00: no auto-logging
- 0x01: auto data-logging if start-up to data stream mode is enabled (STARTUP_MODE[15:0] = 0x01). Refer to the STRMLOG (2Ch) register description for the log file format.

ModbusAddr: 40083~40086, RegisterAddr: 52h~ 55h

Start-up Mode Configuration

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STARTUP_ODR1~4								RW	0x1900

ModbusAddr: 40097~40101, RegisterAddr: 60h~ 64h

Sensor Modbus-TCP Port

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
TCP_P1~4								R	0x01F9 0x01FA 0x01FB 0x01FC 0x01FD

TCPPx, x=1~4 registers contain the Modbus-TCP port number for sensor SNx.This port serves as the gateway that convert between the Modbus-TCP and Modbus-RTU communication protocols to sensor SNx. Below table summarized the default port number:

Sensor Node	Modbus-TCP Port
SN1	505
SN2	506
SN3	507
SN4	508

ModbusAddr: 40102~40105, RegisterAddr: 65h~ 68h

Sensor Stream Port

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
STRM_P1~4								R	0x05E1 0x05E2 0x05E3 0x05E4

STRMPx, x=1~4 registers contain the stream port number for sensor SNx. When stream mode of sensor SNx is enabled by setting STEN_Sx (20h[x])=1’b1,sensor SNx will continuously stream datato Ethernet through this port at the rate set by STRM_ODR [15:0]. Below table summarized the default port number

Sensor Node	Stream Port
SN1	1505
SN2	1506
SN3	1507
SN4	1508

ModbusAddr: 40161~40164, RegisterAddr: A0h~A3h

Hub FW Version

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
VER1~4								R	TBD

Hub FW version information: TBD

ModbusAddr: 40177~40180, RegisterAddr: B0h~B3h

Sensor SN1 FW Version

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
S1_VER1~4								R	N/A

Sensor SN1 FW version information.

ModbusAddr: 40182~40185, RegisterAddr: B5h~ B8h

Sensor SN2 FW Version

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
S2_VER1~4								R	N/A

Sensor SN2 FW version information.

ModbusAddr: 40187~40190, RegisterAddr: BAh~ BDh

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Sensor SN3 FW Version

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
S3_VER1~4								R	N/A

Sensor SN3 FW version information

ModbusAddr: 40192~40195, RegisterAddr: BFh~ C2h

Sensor SN4 FW Version

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	Access	Default
bit7	bit6	bit5	bit4	bit11	bit10	bit1	bit0		
S4_VER1~4								R	N/A

Sensor SN4 FW version information.

5 Modbus Interface

TiHUB-01 provides 5 Modbus-TCP (MBAP) services over the Ethernet: one for the gateway configuration, and 4 for the sensor node bridging.

The Modbus-TCP (MBAP) gateway configuration service is provided through port 502. All availablesetting and gateway information are exposed by registers. Please refer to the "Register Map" sectionfor details.

The Modbus-TCP (MBAP) sensor node bridging services are provided through port 1505~1508. JHN-01 can bridge up to 4 TiACC sensor nodes to Ethernet. The bridging is accomplished by conversion between Modbus-RTU and Modbus-TCP communication protocols. In such task TiHUB-01serves as a Modbus-TCP Server (Slave) that receives request from user Modbus-TCP Client (Master)and convert it to Modbus-RTU request to send to connected TiACC sensor node. In similar way TiHUB-01 get the Modbus-RTU response from TiACC sensor node and convert it to Modbus-TCP response to user Master. User should consult TiACC sensor node datasheet for all available sensornode register settings.

Except explicitly indicated otherwise, TiHUB-01 Modbus-TCP (MBAP) service follows MODBUSstandard as summarized in the below basic rules:

- All communications on the communications loop conforms to a master/slave scheme. TiHUB-01serves as slave in the communication loop.
- The master initiates and controls all information transfer on the communications loop.
- A slave device never initiates a communications sequence.
- All communications activity on the loop occurs in the form of "packets." A packet is a serial string of 8-bit bytes. The maximum number of bytes contained within one request/response packet is 260.
- All packets transmitted by the master are requests. All packets transmitted by a slave device are responses.
- At most one slave can respond to a single request from a master.
- MODBUS uses a 'big-Endian' representation for addresses and data items. This means that when a numerical quantity larger than a single byte is transmitted, the most significant byte is sent first

5.1 Communication Frame Description

Byte	1	2	3	4	5	6	7	8	9	10	…	8+N
Name	TID_H	TID_L	PID_H	PID_L	LEN_H	LEN_L	ADDR	FUNC	DATA1	DATA2	…	DATA{N}

Every Modbus packet consists of six fields:

- Transaction Identifier field: 2 bytes, TID_H and TID_L. Identify of a Modbus request/response transaction.
- Protocol Identifier field: 2 bytes, PID_H and PID_L. Always be 0 for Modbus protocol.
- Length field: 2 bytes, LEN_H and LEN_L.
- Address field: 1 byte, ADDR. For gateway configuration service, use TiHUB-01 slave address,0x41. For sensor node bridging service, use TiACC sensor node slave address.
- Function field: 1 byte, FUNC.
- Data field: multiple N bytes, DATA1, DATA2, … , DATA{N}

5.2 Slave Address

TiHUB-01 slave address is 0x41.

For gateway configuration service, TiHUB-01 performs the command specified in the packet when it receives a request packet with the slave address field matching its own address, 0x41. A response packet generated by TiHUB-01 has the same value, 0x41, in the slave address field.

For sensor node bridging serve, use TiACC sensor node slave address.

5.3 Function Field

TiHUB-01 supports the following Modbus functions:

FunctionCode	Function	Description
01	Read Coil Status	Available for gateway configuration service only. TheON/OFF status of discrete coils of #0 to #7 is mapped toGPIOCONN's DOUT1~8High/Low status respectively.
02	Read Input Status	Available for gateway configuration service only.The ON/OFF status of discrete input of #0 to #7 is mapped toGPIOCONN's DIN1~8 High/Low

		status respectively.
03	Read holding registers	● Gateway configuration service:obtains the current value in one or more holding registers of TiHUB-01. ● Sensor node bridging service:obtains the current valuein one or more holding registers of connected TiACC sensor node.
05	Force Single Coil	Available for gateway configuration service only. Set the ON/OFF status of discrete coils of #0 to #7 is mapped to setGPIOCONN's DOUT1~8 High/Low status respectively.
06	Set single register	● Gatewayconfiguration service: places specific value into a holding register of TiHUB-01. ● Sensor node bridging service: places specific value intoa holding register of connected TiACC sensor node.
15	Force Multiple Coils	Available for ateway configuration service only. Set the ON/OFF status of discrete coils of #0 to #7 is mapped to set GPIOCONN's DOUT1~8 High/Low status respectively.
16	Set multiple registers	● Gateway configuration service: places specific values into a series of consecutive holding registers of TiHUB-01. ● Sensor node bridging service: places specific values intoa series of consecutive holding registers of TiACC sensor node.

The holding registers that can be written to the device are defined in the register map of TiHUB-01 orTiACC sensor node respectively.

If a Modbus master device sends an invalid command to TiHUB-01 or attempts to read an invalid holding register, an exception response is generated. The exception response follows the standard packet format. The high order bit of the function code in an exception response is set to 1. The datafield of an exception response contains the exception error code as show in the follow table.

Exception Code	Name	Description

01	Illegal Function	An invalid command is contained in the function field of the request packet. TiHUB-01 only supports Modbus functions 03, 06 and 16.
02	Illegal Address	The address referenced in the data field is an invalid address for the specified function.
03	Illegal Value	The value referenced in the data field is not allowed for the referred register.

Data Field

The data field of a Modbus request is of variable length, and depends upon the function. This field contains information required by the slave device to perform the command specified in a request packet or data being passed back by the slave device in a response packet.

Data in this field are contained in 16-bit registers. Registers are transmitted in the order of high- byte first, low-byte second, the so called “Big Endian” format.

6 Document History

Revision No.	Description	Date
v0.1.0	Preliminary first release	2021/10/12